Thus, just as in the case of a black hole's center, the early universe falls on both sides of the divide: The enormous density of the early universe requires the use of general relativity. The tiny size of the early universe requires the use of quantum mechanics. But once again, in combination the laws break down. The projector jams, the cosmic film burns up, and we are unable to access the universe's earliest moments. Because of the conflict between general relativity and quantum mechanics, we remain ignorant about what happened at the beginning and are reduced to drawing a fuzzy patch in Figure 10.6.

If we ever hope to understand the origin of the universe—one of the deepest questions in all of science—the conflict between general relativity and quantum mechanics *must* be resolved. We must settle the differences between the laws of the large and the laws of the small and merge them into a single harmonious theory.

The Unlikely Road to a Solution[31]

As the work of Newton and Einstein exemplifies, scientific breakthroughs are sometimes born of a single scientist's staggering genius, pure and simple. But that's rare. Much more frequently, great breakthroughs represent the collective effort of many scientists, each building on the insights of others to accomplish what no individual could have achieved in isolation. One scientist might contribute an idea that sets a colleague thinking, which leads to an observation that reveals an unexpected relationship that inspires an important advance, which starts anew the cycle of discovery. Broad knowledge, technical facility, flexibility of thought, openness to unanticipated connections, immersion in the free flow of ideas worldwide, hard work, and significant luck are all critical parts of scientific discovery. In recent times, there is perhaps no major breakthrough that better exemplifies this than the development of *superstring theory.*

Superstring theory is an approach that many scientists believe successfully merges general relativity and quantum mechanics. And as we will see, there is reason to hope for even more. Although it is still very much a work in progress, superstring theory may well be a fully unified theory of all forces and all matter, a theory that reaches Einstein's